~~H(z) =~~

$$y[n] = \frac{5}{6} y[n-1] - \frac{1}{6} y[n-2] + x[n]$$

Input $x[n] = \delta[n] - \frac{1}{3}\delta[n-1]$

∴ Z transform of the system: [Transfer function]

$$H(z) = \frac{1}{\left[1 - \left(\frac{5}{6}\right)z^{-1} + \frac{1}{6}(z^{-2})\right]}$$

$$= \frac{1}{\left[\left(1 - \frac{z^{-1}}{2}\right)\left(1 - \frac{1}{3}z^{-1}\right)\right]}$$

Z transform of the input:

$$X(z) = 1 - \frac{1}{3}z^{-1}$$

$$Y(z) = X(z) \cdot H(z) = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)}$$

ROC: $|z| > \frac{1}{2}$

($y[n] = 0$ for $n < 0$)

$$y(n) = \left(\frac{1}{2}\right)^n u(n)$$

↳ Casual response $\left(y[n] = 0 \text{ for } n < 0\right)$

$$\sum |y[n]| = 2 \quad (\text{finite}) \rightarrow \text{so response is stable}$$

↳ so response is stable